Multi-Tenant Agentic AI Reference Architecture

Taxonomy, Threat Models & 5-Hop Control Planes on Google Cloud

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Google Cloud

Executive Agenda & Architecture

01 1. Multi-Tenancy Drivers & Taxonomy

02 2. The 5-Hop Agentic Control Plane

03 3. Core Architectural Guardrails

04 4. Threat Vectors & Pattern Trade-Offs

05 5. Production Rollout & Roadmap

1. Multi-Tenancy Drivers & Taxonomy

What is Multi-Tenant Agentic AI?

An architectural framework on Google Cloud that enables sovereign enterprise tenants to run autonomous agents over shared or dedicated resources with cryptographic boundary isolation.

Operates via Google Agent Development Kit (ADK 2.0) + Gemini Enterprise Agent Platform with standard A2A & MCP protocols.

Isolation Vectors

- Tenant Tokens
- Dynamic RLS
- Dedicated CMEK
- Private PSC

Design Tenets

- Zero Data Bleed
- Adaptive Quotas
- Defense-in-Depth
- Production-Verified

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2. The 5-Hop Agentic Control Plane

1

Tenant Ingress

Cloud Armor & IAP authenticate caller claims & enforce WAF rate limits.

2

Context Mediation

AgentGateway extracts tenant ID and mints scoped RFC 8693 authorization tokens.

3

ADK Coordination

Coordinator dispatches user intent to specialist subagents in tenant sandbox.

4

Isolated Execution

Subagent queries Vector Search & BigQuery with tenant CMEK & RLS predicates.

5

Output Synthesis

Sanitization engine scrubs prompt injection echoes and formats final A2UI response.

Five deterministic hops from tenant ingress to cryptographically verified execution

Hops 2 and 3 enforce OIDC token exchange and tenant context attachment before downstream tool dispatch.

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3. Core Architectural Guardrails

1

Workload Identity

SPIFFE/SPIRE workload attestation exchanges client claims for short-lived IAM credentials.

2

Stateless Sandboxing

Cloud Run micro-VM execution guarantees memory isolation between concurrent tenant calls.

3

Cryptographic CMEK

Customer-managed Cloud KMS keys encrypt tenant vectors, tables, and caches independently.

4

Private Networking

Private Service Connect (PSC) eliminates all public internet transit for internal enterprise data.

5

Tamper-Proof Audit

Cloud Logging & BigQuery append-only stores log every agentic decision with tenant tags.

Five defense-in-depth layers enforcing strict multi-tenant boundaries

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Threat Modeling & Blast Radius Matrix

How the architecture mitigates adversarial multi-tenant attack vectors

Prompt Injection Isolation

Contextual boundary firewalls prevent tenant prompts from hijacking shared system prompts.

Noisy Neighbor Defense

Adaptive Cloud Run concurrency limits throttle runaway tenants without impacting neighboring workloads.

Cache Bleed Protection

Vector Search partitions semantic embedding caches strictly by tenant-isolated namespace keys.

Credential Blast Radius

Downstream OAuth 2.0 tokens are scoped to individual operations with minimum TTL (<15 min).

Deterministic Auditing

Tamper-proof BigQuery append-only audit tables record all tool invocations and authorization grants.

Outcome: Guarantees zero cross-tenant contamination across pooled, siloed, and hybrid execution environments.

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4. Threat Vectors & Pattern Trade-Offs

Evaluating Pattern A (Pooled), Pattern B (Siloed), and Pattern C (Hybrid)

Pattern A: Pooled

Shared compute & VPC with logical database RLS.
Lowest operational cost ($0.04/req); ideal for high-volume Tier-1 workloads.

Pattern B: Siloed

Dedicated VPC, Cloud Run service, and dedicated CMEK per tenant.
Maximum compliance isolation for heavily regulated financial and pharma tiers.

Pattern C: Hybrid

Shared gateway control plane with dynamically provisioned sovereign tenant data pods.
Supports zero-downtime tier promotion from pooled to siloed.

Selection Framework

Choose based on regulatory posture, tenant budget constraints, and latency sensitivity requirements.

Operational Uniformity

All three patterns share unified ADK agent code, CI/CD pipelines, and observability schemas.

Architectural Rule: Start high-volume users on Pattern A (Pooled) and provide an automated promotion path to Pattern B (Siloed).

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5. Production Rollout & Roadmap

1

Tenant Classification

Map enterprise business units and data compliance requirements into Pattern A, B, or C tiers.

2

VPC & PSC Network Setup

Establish Private Service Connect endpoints, Cloud Armor security policies, and internal VPC perimeters.

3

Gateway Control Plane

Deploy Cloud Run stateless AgentGateway with RFC 8693 token exchange and OPA Rego policies.

4

Data Plane & CMEK Keys

Provision dedicated Cloud KMS keys and configure BigQuery/Spanner Row-Level Security.

5

Tamper-Proof Audit

Execute automated cross-tenant penetration tests and data bleed verification gates.

6

Canary Traffic Shift

Gradually shift tenant traffic with real-time OpenTelemetry latency and error budget monitoring.

Production Mandate: Six-phase journey from architectural blueprint to multi-tenant production

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Thank you

